Program Development Process

Step 1: Requirements specification

Problem Statement: Create a C++ application that allows a user to log multiple workout sessions (1-10). The program must collect data for each session, calculate fitness metrics (pace and calories), and provide a cumulative final report summarizing the total performance across all sessions.

Step 2: Analyze

- **Inputs**: Number of sessions, distance run (miles), time taken (minutes), and body weight (lbs).
- **Outputs**: Per-session pace and calories burned; total distance, time, and calories; average pace for all sessions.
- **Constants**: Calorie burn coefficient (0.75).

Step 3: Program design

1. Initialize total accumulators to zero.
2. Prompt user for numSessions and validate (1-10).
3. Loop from 1 to numSessions :
    a. Input distance, time, and weight (validate > 0).
    b. Calculate Pace = Time/Distance.
    c. Calculate Calories = 0.75 x Weight x Distance.
    d. Add session values to totals.
4. Calculate final average pace (Total Time/Total Distance).
5. Display the final summary report.

Step 4: Implementation in C++

```
#include <iostream>
#include <iomanip>
#include <cstdlib>

using namespace std;

//Function Prototypes
void getSessionData(int sessionNum, int totalNumSe, double& distance, double& time, double& weight);
double calculatePace(double distance, double time);
double calculateCalories(double weight, double distance);
void updateTotals(double dist, double time, double calories, double& totalDist, double& totalTime,
double& totalCals);
void displaySessionSummary(int sessionNum, double pace, double calories);
void displayFinalReport(int totalSessions, double totalDist, double totalTime, double totalCals);

int main()
{
    //Variable Declarations
    int numSessions;
    double totalDist = 0, totalTime = 0, totalCals = 0;
    double distance, time, weight;
    double currentPace, currentCals;
    char choice;
```

```
    //Set output formatting to 2 decimal places
    cout << fixed << showpoint << setprecision(2);
    do {
        cout << "Welcome to the Fitness Progress Tracker!" << endl;
        //Get and validate the number of sessions to log
        cout << "How many workout sessions would you like to log (1-10)? ";
        cin >> numSessions;

        while (numSessions < 1 || numSessions > 10) {
            cout << "Invalid input. Please enter a number between 1 and 10: ";
            cin >> numSessions;
        }

        cout << endl;
        //Main loop to process each session
        for (int i = 1; i <= numSessions; i++) {
            //Collect user input for the current session
            getSessionData(i, numSessions, distance, time, weight);

            //Perform calculations
            currentPace = calculatePace(distance, time);
            currentCals = calculateCalories(weight, distance);

            //Accumulate totals for the final report
            updateTotals(distance, time, currentCals, totalDist, totalTime, totalCals);

            //Show summary for the individual session
            displaySessionSummary(i, currentPace, currentCals);
            cout << endl;
        }
        //Show the final aggregated results
        displayFinalReport(numSessions, totalDist, totalTime, totalCals);

        cout << "\nWould you like to perform another round of sessions? (Y/N):";
        cin >> choice;
        system("pause");
        system("cls");
    } while (choice == 'Y' || choice == 'y');
    return 0;
}
//Prompts user for session-specific data with input validation loops
void getSessionData(int sessionNum, int totalNumSe, double& distance, double& time, double& weight)
{
    cout << "Session " << sessionNum << " of " << totalNumSe << endl;

    do {
        cout << "Enter distance run (miles): ";
        cin >> distance;
        if (distance <= 0) {
            cout << "Error. Distance must be positive." << endl;
        }
    } while (distance <= 0);

    do {
        cout << "Enter time taken (minutes): ";
        cin >> time;
        if (time <= 0) {
            cout << "Error. Time must be positive." << endl;
        }
    } while (time <= 0);

    do {
        cout << "Enter your body weight (lbs): ";
```

```
        cin >> weight;
        if (weight <= 0) {
            cout << "Error. Weight must be positive." << endl;
        }
    } while (weight <= 0);
}
//Calculates pace in minutes per mile
double calculatePace(double distance, double time) {
    return time / distance;
}
//Calculates calories burned based on a standard MET formula
double calculateCalories(double weight, double distance) {
    return 0.75 * weight * distance;
}
//Adds current session values to the running totals using pass-by-reference
void updateTotals(double dist, double time, double calories, double& totalDist, double& totalTime,
double& totalCals) {
    totalDist += dist;
    totalTime += time;
    totalCals += calories;
}
//Prints a formatted summary for a single session
void displaySessionSummary(int sessionNum, double pace, double calories) {
    cout << "\n----- Session " << sessionNum << " Summary -----" << endl;
    cout << "Pace: \t\t\t" << pace << " min/mile" << endl;
    cout << "Calories Burned: \t" << calories << " cal" << endl;
}
//Prints the final aggregated data for all logged sessions
void displayFinalReport(int totalSessions, double totalDist, double totalTime, double totalCals) {
    double avgPace;
    //Prevent division by zero if no distance was recorded
    if (totalDist > 0) {
        avgPace = totalTime / totalDist;
    }
    else {
        avgPace = 0;
    }

    cout << "\n========= FINAL REPORT =========" << endl;
    cout << "Total Sessions: \t" << totalSessions << endl;
    cout << "Total Distance: \t" << totalDist << " miles" << endl;
    cout << "Total Time: \t\t" << totalTime << " minutes" << endl;
    cout << "Total Calories: \t" << totalCals << " cal" << endl;
    cout << "Average Pace: \t\t" << avgPace << " min/mile" << endl;
}
```

Step 5: Testing

C:\Users\lpant\Documents\CI

```
Welcome to the Fitness Progress Tracker!
How many workout sessions would you like to log (1-10)? -1
Invalid input. Please enter a number between 1 and 10: 11
Invalid input. Please enter a number between 1 and 10: 1

Session 1 of 1
Enter distance run (miles): 3.5
Enter time taken (minutes): 32
Enter your body weight (lbs): 175

----- Session 1 Summary -----
Pace:                    9.14 min/mile
Calories Burned:         459.38 cal

========= FINAL REPORT =========
Total Sessions:          1
Total Distance:          3.50 miles
Total Time:              32.00 minutes
Total Calories:          459.38 cal
Average Pace:            9.14 min/mile

Would you like to perform another round of sessions? (Y/N):
```

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```
Welcome to the Fitness Progress Tracker!
How many workout sessions would you like to log (1-10)? 3

Session 1 of 3
Enter distance run (miles): 3.5
Enter time taken (minutes): 32
Enter your body weight (lbs): 175

----- Session 1 Summary -----
Pace:                    9.14 min/mile
Calories Burned:         459.38 cal

Session 2 of 3
Enter distance run (miles): 2.0
Enter time taken (minutes): 20.5
Enter your body weight (lbs): 175

----- Session 2 Summary -----
Pace:                    10.25 min/mile
Calories Burned:         262.50 cal

Session 3 of 3
Enter distance run (miles): 5.2
Enter time taken (minutes): 45
Enter your body weight (lbs): 175

----- Session 3 Summary -----
Pace:                    8.65 min/mile
Calories Burned:         682.50 cal


========= FINAL REPORT =========
Total Sessions:          3
Total Distance:          10.70 miles
Total Time:              97.50 minutes
Total Calories:          1404.38 cal
Average Pace:            9.11 min/mile

Would you like to perform another round of sessions? (Y/N):
```